

DBSCAN Clustering

Albert Y. Kim

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) is a machine learning algorithm that groups data points based on how closely packed they are, i.e. how dense they are, allowing it to find clusters of any shape while ignoring outliers.

DBSCAN

DBSCAN finds clusters using two parameters:

1. a radius `eps`
2. a minimum number of data-points `min_samples`

DBSCAN

DBSCAN finds clusters using two parameters:

1. a radius `eps`
2. a minimum number of data-points `min_samples`

The algorithm surrounds each data-point with a *hypersphere* (aka circle if $p = 2$ dimensions). These hyperspheres are used to label each data-point as one of three types:

1. *Core points* contain at least `min_samples` neighbors within their hypersphere
2. *Border aka boundary points* contain at least 1 neighbor within their hypersphere
3. *Noise points* are at least `eps` away from any other data-point

DBSCAN

Cluster membership are then determined by connected density regions. The diagram below illustrates DBSCAN with `min_samples = 3`:

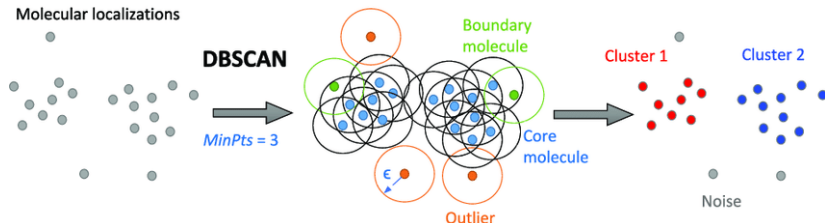
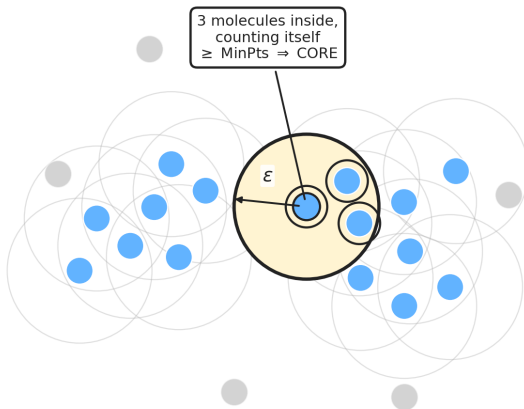


Image Source: [A Review of Super-Resolution Single-Molecule Localization Microscopy Cluster Analysis and Quantification_Methods](#)

DBSCAN: Breaking Down Middle Image

1. Core molecules

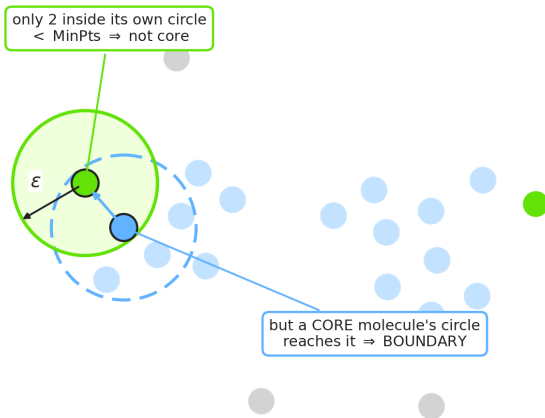
A CORE molecule has at least $\text{MinPts} = 3$ molecules inside its ϵ -circle, counting itself. These are the molecules in the dense interior.



DBSCAN: Middle Image 2

2. Boundary molecules

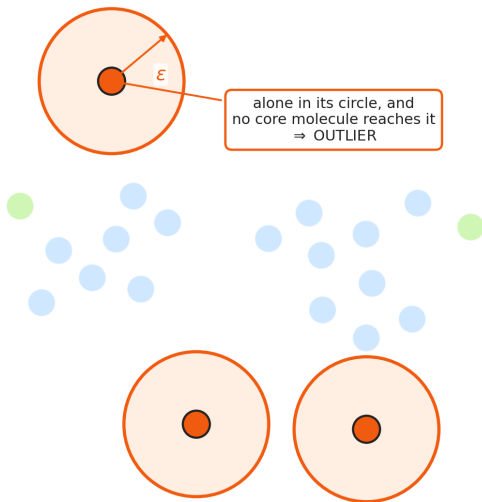
A BOUNDARY molecule has fewer than $\text{MinPts} = 3$ molecules in its own ϵ -circle, but it sits inside the ϵ -circle of a core molecule.



DBSCAN: Middle Image 3

3. Outlier molecules

An OUTLIER has fewer than $\text{MinPts} = 3$ molecules in its ϵ -circle, and sits inside no core molecule's ϵ -circle. Nothing reaches it, so it joins no cluster.



Outlier (anomaly) detection

- ▶ In statistics, an outlier is a data-point that is significantly far from other observations
 - ▶ A simple example is the “3 sigma rule”, which will classify anything more than 3 standard deviations from the mean as an outlier
 - ▶ This amounts most extreme 0.3% of data-points under a normal model
- ▶ DBSCAN provides a flexible method of outlier detection by **tuning** the eps hyperparameter
 - ▶ This can be set using domain-specific knowledge, or tuned so that a certain percentage of the data is classified as outliers

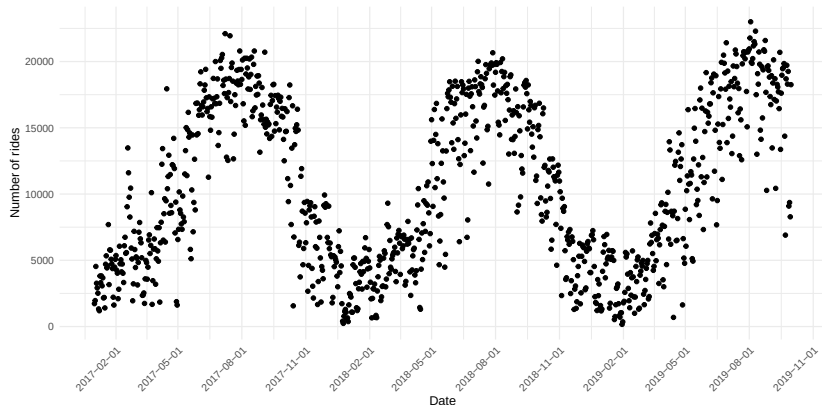
Ex: Chicago Divvy Bikeshare Data

From `data/rides.csv`: Data of number of rides between between 2017-01-01 and 2019-09-30 ($n = 1003$). Here are the first 5 rows:

day	num_rides
2017-01-01	1727
2017-01-02	1960
2017-01-03	4537
2017-01-04	3269
2017-01-05	2917
2017-01-06	2516

Ex: Time Series Plot of Data

We plot the number of rides on the y-axis:



Ex: Apply DBSCAN

After

- ▶ Converting dates 2017-01-01, 2017-01-02, ..., 2019-09-30 to integers 1, 2, ..., 1003
- ▶ Standardizing y the number of rides

we apply DBSCAN on this $p = 2$ dimensional data set with `min_samples = 2` and:

1. `eps = 1.6`
2. `eps = 1.3`

Ex: Chicago Divvy bikeshare data

Note how for larger eps fewer points are labeled as outliers

